

Ansible-03

Tasks

1. Write a single ansible playbook which will install apache and nginx. Note: Playbook should not be hardcoded and pass the variables from different file.
2. Ansible playbook to create 10 different directories with minimal code and directory names should be passed as variables.
3. Ansible playbook to copy ssh-keygen from master to worker nodes. Note: a)Provision new 3 ec2 machines, one master and two worker nodes. b)Create common user called ansadm and provide sudo privileges on 3 ec2 instances. c)Create ssh-keygen in master and your playbook should copy the keygen making it password less authentication.
4. Ansible playbook to inject ansible vault variables.

Task 1:

1. Write a single ansible playbook which will install apache and nginx. Note: Playbook should not be hardcoded and pass the variables from different file.

Step:1

Create two files one for variables and other for playbookuse

```
root@ip-172-31-2-72:~/playbooks# vi web.yml
root@ip-172-31-2-72:~/playbooks# vi var.yml
```

Step2 : Inside web.yml

```
- name: Install and configure Apache & Nginx
  hosts: all
  become: yes
  vars_files:
    - var.yml

  tasks:
    # -----
    # Update & Install packages
    # -----
    - name: Update apt cache
      apt:
        update_cache: yes

    - name: Install Apache
      apt:
        name: apache2
        state: present

    - name: Install Nginx
      apt:
        name: nginx
        state: present

    # -----
    # Configure Apache
    # -----
    - name: Set Apache listen port
      lineinfile:
        path: /etc/apache2/ports.conf
        regexp: '^Listen'
        line: "Listen {{ apache_port }}"
        notify: Restart Apache

    - name: Configure Apache vhost
      copy:
        dest: /etc/apache2/sites-enabled/000-default.conf
        content: |
          <VirtualHost *:{{ apache_port }}>
            DocumentRoot /var/www/html
```

i-08544442c9a14b500 (ansible-master)

PublicIPs: 13.63.53.133 PrivateIPs: 172.31.2.72

```
- name: Install and configure Apache & Nginx
  hosts: all
  become: yes
  vars_files:
    - vars.yml
```

tasks:

```
# -----
# Update & Install packages
# -----
```

```
- name: Update apt cache
  apt:
    update_cache: yes
```

```
- name: Install Apache
  apt:
    name: apache2
    state: present
```

```
- name: Install Nginx
  apt:
    name: nginx
    state: present
```

```
# -----
# Configure Apache
# -----
```

```
- name: Set Apache listen port
  lineinfile:
    path: /etc/apache2/ports.conf
    regexp: '^Listen'
    line: "Listen {{ apache_port }}"
  notify: Restart Apache
```

```
- name: Configure Apache vhost
  copy:
    dest: /etc/apache2/sites-enabled/000-default.conf
    content: |
      <VirtualHost *:{{ apache_port }}>
        DocumentRoot /var/www/html
        <Directory /var/www/html>
          AllowOverride All
          Require all granted
        </Directory>
```

</VirtualHost>

notify: Restart Apache

Configure Nginx (Ubuntu safe)

- name: Disable default nginx site (port 80)

file:

path: /etc/nginx/sites-enabled/default

state: absent

notify: Restart Nginx

- name: Configure Nginx on port 81

copy:

dest: /etc/nginx/conf.d/nginx81.conf

content: |

server {

listen {{ nginx_port }};

server_name _;

root /usr/share/nginx/html;

index index.html;

}

notify: Restart Nginx

Start and enable services

- name: Start and enable Apache

systemd:

name: apache2

state: started

enabled: yes

- name: Start and enable Nginx

systemd:

name: nginx

state: started

enabled: yes

handlers:

- name: Restart Apache

service:

name: apache2

state: restarted

- name: Restart Nginx service:
 - name: nginx
 - state: restarted

--> save and exit
Now inside

var.yml file:

```
apache_port: 80
nginx_port: 81
```

Save and exit

--> step 3:
Now check the

--> `ansible-playbook web.yum --syntax-check`

—-> now run the playbook

—> ansible-playbook web.yml

```
root@ip-172-31-2-72:~/playbooks# ansible-playbook web.yml

PLAY [Install and configure Apache & Nginx] *****

TASK [Gathering Facts] *****
ok: [172.31.33.34]
ok: [172.31.37.159]

TASK [Update apt cache] *****
changed: [172.31.33.34]
changed: [172.31.37.159]

TASK [Install Apache] *****
ok: [172.31.37.159]
ok: [172.31.33.34]

TASK [Install Nginx] *****
ok: [172.31.37.159]
ok: [172.31.33.34]

TASK [Set Apache listen port] *****
ok: [172.31.37.159]
ok: [172.31.33.34]

TASK [Configure Apache vhost] *****
ok: [172.31.33.34]
ok: [172.31.37.159]

TASK [Disable default nginx site (port 80)] *****
ok: [172.31.33.34]
ok: [172.31.37.159]

TASK [Configure Nginx on port 81] *****
ok: [172.31.33.34]
ok: [172.31.37.159]

TASK [Start and enable Apache] *****
ok: [172.31.37.159]
ok: [172.31.33.34]

TASK [Start and enable Nginx] *****
```

i-08544442c9a14b500 (ansible-master)

PublicIPs: 13.63.53.133 PrivateIPs: 172.31.2.72

```
ok: [172.31.37.159]

TASK [Disable default nginx site (port 80)] *****
ok: [172.31.33.34]
ok: [172.31.37.159]

TASK [Configure Nginx on port 81] *****
ok: [172.31.33.34]
ok: [172.31.37.159]

TASK [Start and enable Apache] *****
ok: [172.31.37.159]
ok: [172.31.33.34]

TASK [Start and enable Nginx] *****
ok: [172.31.33.34]
ok: [172.31.37.159]

PLAY RECAP *****
172.31.33.34      : ok=10  changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.37.159    : ok=10  changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

—> now go and check your worker1 and worker2
PublicIP:80,81

```
root@ip-172-31-33-34:~# systemctl status apache2
Unit apache2.service could not be found.
root@ip-172-31-33-34:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-01-30 09:46:08 UTC; 8min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 3161 (apache2)
    Tasks: 55 (limit: 1017)
   Memory: 6.5M (peak: 6.7M)
      CPU: 55ms
   CGroup: /system.slice/apache2.service
           └─3161 /usr/sbin/apache2 -k start
             └─3163 /usr/sbin/apache2 -k start
               └─3164 /usr/sbin/apache2 -k start

Jan 30 09:46:08 ip-172-31-33-34 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 30 09:46:08 ip-172-31-33-34 systemd[1]: Started apache2.service - The Apache HTTP Server.
root@ip-172-31-33-34:~# cd /var/www/html/
root@ip-172-31-33-34:/var/www/html# ls
index.html  index.nginx-debian.html
root@ip-172-31-33-34:/var/www/html# cat index.html
hello white tiger
root@ip-172-31-33-34:/var/www/html# cd
root@ip-172-31-33-34:~# systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-01-30 09:58:24 UTC; 19min ago
     Docs: man:nginx(8)
   Process: 5886 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 5888 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 5889 (nginx)
    Tasks: 3 (limit: 1017)
   Memory: 2.5M (peak: 2.9M)
      CPU: 19ms
   CGroup: /system.slice/nginx.service
           └─5889 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─5890 "nginx: worker process"
               └─5891 "nginx: worker process"

Jan 30 09:58:24 ip-172-31-33-34 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jan 30 09:58:24 ip-172-31-33-34 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
```

i-0e57dafb81ac0e460 (worker1)

PublicIPs: 13.60.170.157 PrivateIPs: 172.31.33.34

```
root@ip-172-31-37-159:~# systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-01-30 09:58:24 UTC; 20min ago
     Docs: man:nginx(8)
   Process: 5729 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 5731 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 5732 (nginx)
    Tasks: 3 (limit: 1017)
   Memory: 2.4M (peak: 2.9M)
      CPU: 22ms
   CGroup: /system.slice/nginx.service
           └─5732 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─5733 "nginx: worker process"
               └─5734 "nginx: worker process"

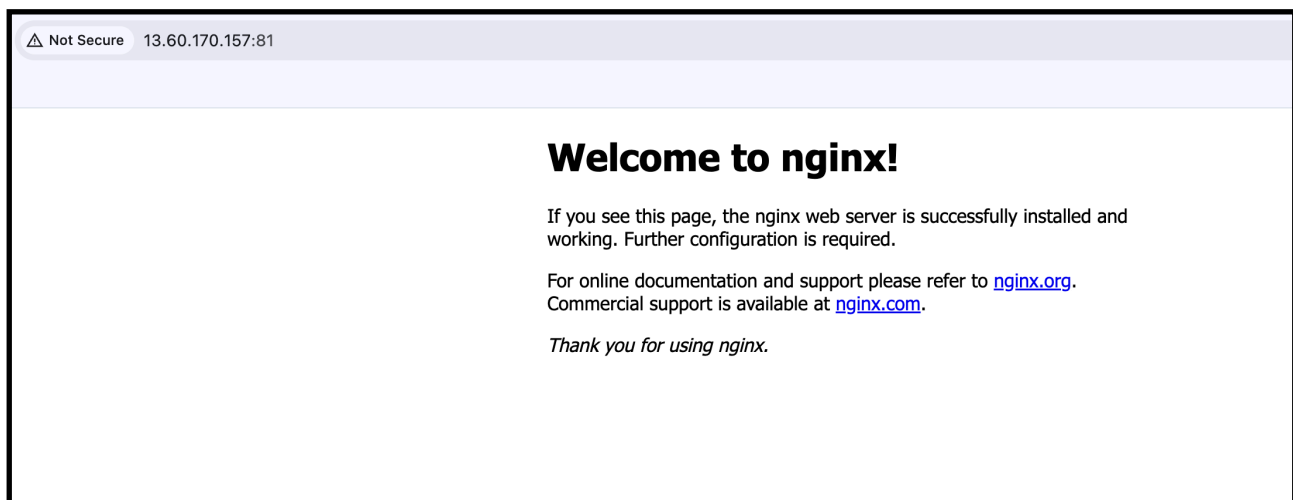
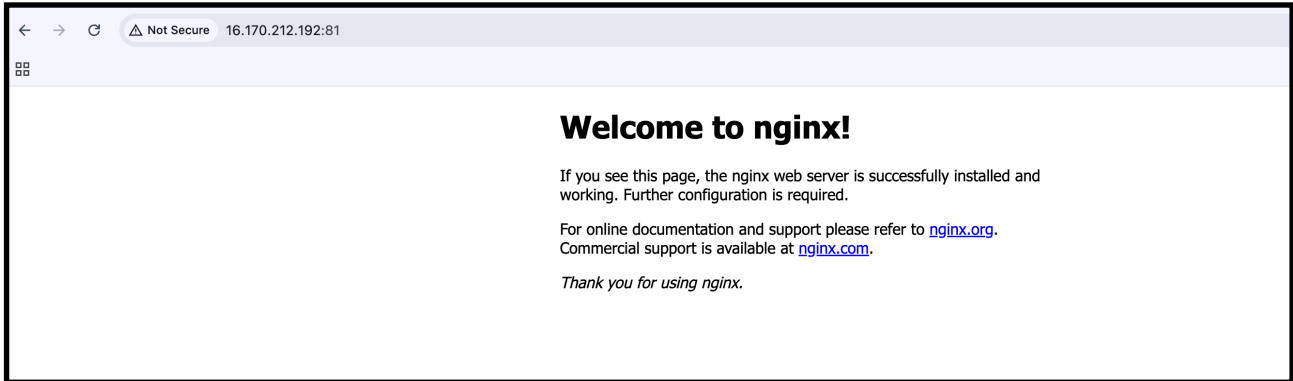
Jan 30 09:58:24 ip-172-31-37-159 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jan 30 09:58:24 ip-172-31-37-159 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
root@ip-172-31-37-159:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-01-30 09:58:24 UTC; 20min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 5643 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 5646 (apache2)
    Tasks: 55 (limit: 1017)
   Memory: 6.4M (peak: 6.9M)
      CPU: 98ms
   CGroup: /system.slice/apache2.service
           └─5646 /usr/sbin/apache2 -k start
             └─5648 /usr/sbin/apache2 -k start
               └─5649 /usr/sbin/apache2 -k start

Jan 30 09:58:24 ip-172-31-37-159 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 30 09:58:24 ip-172-31-37-159 systemd[1]: Started apache2.service - The Apache HTTP Server.
root@ip-172-31-37-159:~#
```

i-09212a024d7a75d98 (worker2)

PublicIPs: 16.170.212.192 PrivateIPs: 172.31.37.159

—> check ip in both servers



Task 2 :

2. Ansible playbook to create 10 different directories with minimal code and directory names should be passed as variables.

Inside playbook directory

- > create file name = dirs.yml
- > now add below add code and save
- > :wq!

```
---
- name: Create multiple directories using variables
  hosts: all
  become: true

  vars:
    dir_list:
      - /opt/app1
      - /opt/app2
      - /opt/app3
      - /opt/app4
      - /opt/app5
      - /opt/app6
      - /opt/app7
      - /opt/app8
      - /opt/app9
      - /opt/app10

  tasks:
    - name: Create directories
      file:
        path: "{{ item }}"
        state: directory
        mode: '0755'
        loop: "{{ dir_list }}"
```

- > ansible-playbook dirs.yml --syntax-check
- > ansible-playbook dirs.yml

```

root@ip-172-31-2-72:~/playbooks# vi dirs.yml
root@ip-172-31-2-72:~/playbooks# ansible-playbook dirs.yml --syntax-check

playbook: dirs.yml
root@ip-172-31-2-72:~/playbooks# ansible-playbook dirs.yml

PLAY [Create multiple directories using variables] *****

TASK [Gathering Facts] *****
ok: [172.31.37.159]
ok: [172.31.33.34]

TASK [Create directories] *****
changed: [172.31.37.159] => (item=/opt/app1)
changed: [172.31.33.34] => (item=/opt/app1)
changed: [172.31.37.159] => (item=/opt/app2)
changed: [172.31.33.34] => (item=/opt/app2)
changed: [172.31.37.159] => (item=/opt/app3)
changed: [172.31.33.34] => (item=/opt/app3)
changed: [172.31.37.159] => (item=/opt/app4)
changed: [172.31.33.34] => (item=/opt/app4)
changed: [172.31.37.159] => (item=/opt/app5)
changed: [172.31.33.34] => (item=/opt/app5)
changed: [172.31.37.159] => (item=/opt/app6)
changed: [172.31.33.34] => (item=/opt/app6)
changed: [172.31.37.159] => (item=/opt/app7)
changed: [172.31.33.34] => (item=/opt/app7)
changed: [172.31.37.159] => (item=/opt/app8)
changed: [172.31.33.34] => (item=/opt/app8)
changed: [172.31.37.159] => (item=/opt/app9)
changed: [172.31.33.34] => (item=/opt/app9)
changed: [172.31.37.159] => (item=/opt/app10)
changed: [172.31.33.34] => (item=/opt/app10)

PLAY RECAP *****
172.31.33.34      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.37.159   : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

root@ip-172-31-2-72:~/playbooks# vi dirs.yml
root@ip-172-31-2-72:~/playbooks#

```

i-08544442c9a14b500 (ansible-master)

—-> go and check the path cd/opt

—-> and ls

We will able to see all directories

```

root@ip-172-31-33-34:/opt# ls
tomcat
root@ip-172-31-33-34:/opt# ls
app1 app10 app2 app3 app4 app5 app6 app7 app8 app9 tomcat
root@ip-172-31-33-34:/opt#

```

i-0e57dafb81ac0e460 (worker1)

PublicIPs: 13.60.170.157 PrivateIPs: 172.31.33.34

```

root@ip-172-31-37-159:~# cd /opt
root@ip-172-31-37-159:/opt# ls
app1 app10 app2 app3 app4 app5 app6 app7 app8 app9 tomcat
root@ip-172-31-37-159:/opt#

```

i-09212a024d7a75d98 (worker2)

PublicIPs: 16.170.212.192 PrivateIPs: 172.31.37.159

Task 3:

3. Ansible playbook to copy ssh-keygen from master to worker nodes. Note:

- a) Provision new 3 ec2 machines, one master and two worker nodes.
- b) Create common user called **ansadm** and provide sudo privileges on 3 ec2 instances.
- c) Create ssh-keygen in master and your playbook should copy the keygen making it password less authentication.

Requirements:

- a) Provision **3 EC2 machines** (1 master + 2 workers)
- b) Create common user **ansadm** with sudo on all 3 EC2s
- c) Generate **SSH key on master** and copy it to workers for **passwordless authentication**

a) Provision new 3 ec2 machines, one master and two worker nodes.

--> create a file name master-worker.yml

```
root@ip-172-31-1-201:~/playbooks# vi master-workers.yml
```

```

---
- name: Provision EC2s (Master + Workers)
  hosts: localhost
  connection: local
  gather_facts: no

  vars:
    region: eu-north-1
    key_name: ansible
    instance_type: t3.micro
    ami_id: ami-04233b5aecce09244

  tasks:
    - name: Launch Master EC2
      amazon.aws.ec2_instance:
        name: ansible-master
        region: "{{ region }}"
        image_id: "{{ ami_id }}"
        instance_type: "{{ instance_type }}"
        key_name: "{{ key_name }}"
        wait: yes
        network:
          assign_public_ip: true
        tags:
          Role: master

    - name: Launch Worker EC2s
      amazon.aws.ec2_instance:
        name: ansible-worker
        region: "{{ region }}"
        image_id: "{{ ami_id }}"
        instance_type: "{{ instance_type }}"
        key_name: "{{ key_name }}"
        count: 2
        wait: yes
        network:
          assign_public_ip: true
        tags:
          Role: worker

```

"master-workers.yml" 40L, 928B

—> save exit

—> ansible-playbook master-workers.yml

```
root@ip-172-31-1-201:~/playbooks# ansible-playbook master-workers.yml

PLAY [Provision EC2s (Master + Workers)] *****
TASK [Launch Master EC2] *****
changed: [localhost]
TASK [Launch Worker EC2s] *****
changed: [localhost]
PLAY RECAP *****
localhost : ok=2    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

root@ip-172-31-1-201:~/playbooks# vi master-workers.yml
```

☐	ansible-worker	i-0250553e25c582d4c	Running	t3.micro	Initializing	View alarms +	eu-north-1a	ec2-51-20-249-116.eu-...
☐	ansible-worker	i-02c1d78e5dce74f4a	Running	t3.micro	Initializing	View alarms +	eu-north-1a	ec2-13-51-55-255.eu-n...
☐	ansible-master	i-0bf140b006b693e0e	Running	t3.micro	Initializing	View alarms +	eu-north-1a	ec2-13-61-142-112.eu-...
3 instances selected								

b) Create common user called ansadm and provide sudo privileges on 3 ec2 instances

- > now go to your newly created ansible-master
- > install ansible in it

copy your pem key to the controller node and give chmod400
after that give the path of the key in your host file

```
[ec2-user@ip-172-31-25-151 ~]$ vi /home/ec2-user/key.pem
[ec2-user@ip-172-31-25-151 ~]$ chmod 400 /home/ec2-user/key.pem
```

```
[all]
172.31.25.151 ansible_user=ec2-user ansible_ssh_private_key_file=/home/ec2-user/key.pem
172.31.21.111 ansible_user=ec2-user ansible_ssh_private_key_file=/home/ec2-user/key.pem
172.31.30.253 ansible_user=ec2-user ansible_ssh_private_key_file=/home/ec2-user/key.pem

[master]
172.31.25.151

[workers]
172.31.21.111
172.31.30.253
```

```
[ec2-user@ip-172-31-25-151 ~]$ sudo vi /etc/ansible/hosts
[ec2-user@ip-172-31-25-151 ~]$ sudo cat /etc/ansible/hosts
[all]
172.31.25.151 ansible_user=ec2-user ansible_ssh_private_key_file=/home/ec2-user/key.pem
172.31.21.111 ansible_user=ec2-user ansible_ssh_private_key_file=/home/ec2-user/key.pem
172.31.30.253 ansible_user=ec2-user ansible_ssh_private_key_file=/home/ec2-user/key.pem

[master]
172.31.25.151

[workers]
172.31.21.111
172.31.30.253

[ec2-user@ip-172-31-25-151 ~]$
```

Ansible needs a way to SSH into each node.

```
ec2-user@ip-172-31-25-151 ansible$ ansible all -m ping
The authenticity of host '172.31.21.111 (172.31.21.111)' can't be established.
ED25519 key fingerprint is SHA256:Fm+6JaTBVzwb4LsP350lTIsVMMQqInVzuEIn0S7uM.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? [WARNING]: Platform linux on host 172.31.30.253 is using the discovered Python interpreter at /usr/bin/python3.9, but f
uture installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.30.253 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Platform linux on host 172.31.25.151 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.25.151 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
ansible all -m ping
Please type 'yes', 'no' or the fingerprint: yes
[WARNING]: Platform linux on host 172.31.21.111 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.21.111 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
ec2-user@ip-172-31-25-151 ansible$ cd
```

Since AWS EC2 instances usually only allow key-based login, you provide the **private key path** on the machine where you run Ansible (usually the master).

Create a `ansadm.yml` playbook and run

```
[ec2-user@ip-172-31-25-151 ~]$ vi ansadm.yml
[ec2-user@ip-172-31-25-151 ~]$ ansible-playbook ansadm.yml --check

PLAY [Create ansadm user on all nodes] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 172.31.21.111 is using the discovered Python interpreter at /usr/bin/python3.9, but fu
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for
ok: [172.31.21.111]
[WARNING]: Platform linux on host 172.31.25.151 is using the discovered Python interpreter at /usr/bin/python3.9, but fu
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for
ok: [172.31.25.151]
```

```
---
- name: Create ansadm user on all nodes
  hosts: all
  become: yes

  tasks:
    - name: Ensure ansadm user exists
      user:
        name: ansadm
        shell: /bin/bash
        create_home: yes

    - name: Give ansadm sudo privileges without password
      copy:
        dest: /etc/sudoers.d/ansadm
        content: "ansadm ALL=(ALL) NOPASSWD:ALL\n"
        mode: '0440'
```

—> ansible-playbook ansadm.yml

```
[ec2-user@ip-172-31-25-151 ~]$ ansible-playbook ansadm.yml

PLAY [Create ansadm user on all nodes] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 172.31.21.111 is using the discovered Python interpreter at /usr/bin/python3.9, but future install
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more info
ok: [172.31.21.111]
[WARNING]: Platform linux on host 172.31.25.151 is using the discovered Python interpreter at /usr/bin/python3.9, but future install
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more info
ok: [172.31.25.151]
[WARNING]: Platform linux on host 172.31.30.253 is using the discovered Python interpreter at /usr/bin/python3.9, but future install
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more info
ok: [172.31.30.253]

TASK [Ensure ansadm user exists] *****
changed: [172.31.21.111]
changed: [172.31.30.253]
changed: [172.31.25.151]

TASK [Give ansadm sudo privileges without password] *****
changed: [172.31.21.111]
changed: [172.31.25.151]
changed: [172.31.30.253]

PLAY RECAP *****
172.31.21.111      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.25.151      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.30.253      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Process explanation:

1. **Target all nodes:** hosts: all means the playbook runs on **master + worker nodes**.
2. **Use sudo privileges:** become: yes allows creating users and modifying system files.
3. **Create the user:.**

At the end of this step, all nodes have the same ansadm user.

Give ansadm sudo privileges

1. Creates a file /etc/sudoers.d/ansadm on each node.
2. Adds the line:
3. ansadm ALL=(ALL) NOPASSWD:ALL
 - This allows ansadm to run **any command with sudo without entering a password**.
4. mode: '0440' sets proper file permissions for security (only root can read).

At the end of this step, ansadm can run administrative commands on any node.

Sudo su – ansadm on master node

Break it into pieces

- **sudo** → run command as root
- **su** → switch user
- **- ansadm** → switch to user **ansadm** *with full login environment*

So this means:

“Become the ansadm user using root privileges.”

```
'_#_
~\###_ Amazon Linux 2023
~~\#####\
~~\###|
~~\#/ https://aws.amazon.com/linux/amazon-linux-
~~v~'-'->
~~~~
~~._. /
~/m/' -
```

Last login: Fri Jan 30 12:53:03 2026 from 172.31.25.151
[ec2-user@ip-172-31-21-111 ~]\$ sudo su - ansadm
[ansadm@ip-172-31-21-111 ~]\$ ll
total 0

—> above is worker1

—> below is worker 2

```
'\_#  
~\_##### Amazon Linux 2023  
~~\_#####\  
~~\_###|  
~~\_#/ https://aws.amazon.com/linux/amazon-linux-2023  
~~V~'-'->  
~~~~  
~~.-.  
~~\_/  
~~\_m/'
```

Last login: Fri Jan 30 12:53:03 2026 from 172.31.25.151
[ec2-user@ip-172-31-30-253 ~]\$ sudo su - ansadm
[ansadm@ip-172-31-30-253 ~]\$ █

```
-bash: cd: sudoers.d/: permission denied
[ec2-user@ip-172-31-25-151 etc]$ sudo su - ansadm
[ansadm@ip-172-31-25-151 ~]$
```

i-0bf140b006b693e0e (ansible-master)

PublicIPs: 13.61.142.112 PrivateIPs: 172.31.25.151

This command is used to **VERIFY** that:

1. `ansadm` user exists
2. `ansadm` can become root **without password**

If `sudo` was **NOT** configured correctly, this command would:

- ask for a password ❌
- or fail ❌

c) Create ssh-keygen in master and your playbook should copy the keygen making it password less authentication.

Objective

To create SSH key on the **master node** and copy the public key to **worker nodes** so that passwordless authentication is enabled.

Steps Performed

1. Used the master node as the Ansible controller
 - Master node runs Ansible.
 - Worker nodes are managed by the master using private IPs.
2. Ensured **ansadm** user already exists on all nodes
 - This was completed in **Task 3 – Part (b)**.
 - **ansadm** has sudo privileges on master and workers.
- 3.

--> vi ssh-key-copy.yml

```
ansadm.yml key.pem
[ec2-user@ip-172-31-25-151 ~]$ cd ~
vi ssh-key-copy.yml
[ec2-user@ip-172-31-25-151 ~]$ ansible-playbook ssh-key-copy.yml --check
```

```
--
- name: Generate SSH key on master for ec2-user
  hosts: master
  become: yes
  vars:
    ssh_key_path: /home/ec2-user/.ssh/id_rsa
  tasks:
    - name: Ensure .ssh directory exists for ec2-user
      file:
        path: /home/ec2-user/.ssh
        state: directory
        owner: ec2-user
        group: ec2-user
        mode: '0700'

    - name: Generate SSH key pair if not exists
      openssh_keypair:
        path: "{{ ssh_key_path }}"
        type: rsa
        size: 4096
        state: present
        owner: ec2-user
        group: ec2-user
        mode: '0600'

- name: Copy master's public key to ansadm on all workers
  hosts: workers
  become: yes
  vars:
    master_public_key_path: /home/ec2-user/.ssh/id_rsa.pub
  tasks:
    - name: Ensure ansadm exists
      user:
        name: ansadm
        shell: /bin/bash
        create_home: yes

    - name: Ensure .ssh directory exists for ansadm
      file:
        path: /home/ansadm/.ssh
        state: directory
```

i-0bf140b006b693e0e (ansible-master)

PublicIPs: 13.61.142.112 PrivateIPs: 172.31.25.151

- name: Generate SSH key on master for ec2-user
 - hosts: master
 - become: yes
 - vars:
 - ssh_key_path: /home/ec2-user/.ssh/id_rsa
 - tasks:
 - name: Ensure .ssh directory exists for ec2-user
 - file:
 - path: /home/ec2-user/.ssh
 - state: directory
 - owner: ec2-user
 - group: ec2-user
 - mode: '0700'
 - name: Generate SSH key pair if not exists
 - openssh_keypair:
 - path: "{{ ssh_key_path }}"
 - type: rsa
 - size: 4096
 - state: present
 - owner: ec2-user
 - group: ec2-user
 - mode: '0600'
- name: Copy master's public key to ansadm on all workers
 - hosts: workers
 - become: yes
 - vars:
 - master_public_key_path: /home/ec2-user/.ssh/id_rsa.pub
 - tasks:
 - name: Ensure ansadm exists
 - user:
 - name: ansadm
 - shell: /bin/bash
 - create_home: yes
 - name: Ensure .ssh directory exists for ansadm
 - file:
 - path: /home/ansadm/.ssh
 - state: directory
 - owner: ansadm
 - group: ansadm
 - mode: '0700'
 - name: Copy master's public key to ansadm authorized_keys
 - authorized_key:
 - user: ansadm
 - state: present
 - key: "{{ lookup('file', master_public_key_path) }}"

Created **.ssh** directory for **ec2-user** on the master node

- Path: `/home/ec2-user/.ssh`
- Permissions set to `700` for security.

Generated SSH key pair on the master node using Ansible

- Used `openssh_keypair` module.
- Key path:
 - Private key: `/home/ec2-user/.ssh/id_rsa`
 - Public key: `/home/ec2-user/.ssh/id_rsa.pub`
- Key is generated **only if it does not already exist**.

Created **.ssh** directory for **ansadm** on worker nodes

- Path: `/home/ansadm/.ssh`
- Ownership set to `ansadm`.
- Permissions set to `700`.

Copied master's public key to worker nodes

- Used `authorized_key` module.
- Public key (`id_rsa.pub`) from master is added to:

This allows the master to SSH into workers as `ansadm` without password.

--> now run the workspace

Ansible-playbook `ssh-key-copy.yml`

Worker2

```
[ec2-user@ip-172-31-25-151 ~]$ ssh ansadm@172.31.21.111  
#  
_\\_##### Amazon Linux 2023  
~~~\\_#####\n~~~~\###|\n~~~~\##/\n~~~~V~'-'->\n~~~~\n~~~~.\n~~~~./\n~~~~/_m/'-\n\nLast login: Fri Jan 30 13:18:27 2026  
[ansadm@ip-172-31-21-111 ~]$ exit  
logout  
Connection to 172.31.21.111 closed.  
[ec2-user@ip-172-31-25-151 ~]$ vi ssh-key-copy.yml  
[ec2-user@ip-172-31-25-151 ~]$ ssh ansadm@172.31.30.253  
#  
_\\_##### Amazon Linux 2023  
~~~\\_#####\n~~~~\###|\n~~~~\##/\n~~~~V~'-'->\n~~~~\n~~~~.\n~~~~./\n~~~~/_m/'-\n\nLast login: Fri Jan 30 13:16:46 2026  
[ansadm@ip-172-31-30-253 ~]$
```

i-0bf140b006b693e0e (ansible-master)

PublicIPs: 13.61.142.112 PrivateIPs: 172.31.25.151

Task 4:

4. Ansible playbook to inject ansible vault variables.

What is Ansible Vault?

Ansible Vault is used to **encrypt sensitive data** such as:

- passwords
- secrets
- tokens
- API keys

So they are **not visible** in plain text inside playbooks.

Meaning of “inject Ansible Vault variables”

It means:

Store sensitive variables in an **encrypted file** and **use (inject) them inside a playbook** at runtime.

vault variables.

1) create a file name vault.yml

```
root@ip-172-31-1-201:~/playbooks# vi vault.yml
root@ip-172-31-1-201:~/playbooks# cat vault.yml
db_user: admin
db_password: admin@123
```

Add the data

```
db_user: admin
db_password: admin@123
```


From the same directory:

```
ansible-vault encrypt vault.yml
```

You will be asked:

New Vault password:

Confirm New Vault password:

After this, check the file:

```
cat vault.yml
```

Output will look like this:

```
root@ip-172-31-1-201:~/playbooks# ansible-vault encrypt vault.yml
New Vault password:
Confirm New Vault password:
Encryption successful
root@ip-172-31-1-201:~/playbooks# cat vault.yml
$ANSIBLE_VAULT;1.1;AES256
31313937346138363961616535613631363033656335383538656133323062313934383465393733
6566353736373336643039316435336336663363623163350a346335653765333131383861393732
32643033323464333564383663333930373166366234656464353936376230666265383938333134
3562363765346365390a353937313235356630363161653737633264623935643739616634356564
35376163303235623730373163613032633839646561386462333761356133366338633461316263
3534393762373738646237666531636138336532383535303165
```

Since it's encrypted, you **cannot** view it with `cat` or `vi` directly anymore.

Here are the **correct ways to view/edit vault data**

Method 1: View vault contents (READ-ONLY)

```
ansible-vault view vault.yml
```

It will ask:

Vault password:

After you enter the password, you'll see:

```
root@ip-172-31-1-201:~/playbooks# ansible-vault view vault.yml
Vault password:
db_user: admin
db_password: admin@123
```

```
db_user: admin
db_password: admin@123
```

Method 2: Edit vault contents (RECOMMENDED)

```
ansible-vault edit vault.yml
```

- Asks for vault password
- Opens file in editor

```
root@ip-172-31-1-201:~/playbooks# ansible-vault edit vault.yml
Vault password:
root@ip-172-31-1-201:~/playbooks#
```

- Lets you modify values
- Re-encrypts automatically on save

```
db_user: admin
db_password: admin@123
```

```
~
~
~
~
~
~
~
```

2) Create playbook to INJECT vault variables

- > vault-playbook.yml
- > write below code

```
root@ip-172-31-1-201:~/playbooks# vi vault-playbook.yml
```

```
---
- name: Inject Ansible Vault variables
  hosts: localhost
  gather_facts: no
  vars_files:
    - vault.yml

  tasks:
    - name: Use vault variables
      debug:
        msg:
          - "DB User is {{ db_user }}"
          - "DB Password is {{ db_password }}"

~
~
~
~
~
```

—> save :wq!

—>vars_files is how vault variables are injected.

—> we have used vault.yml as vars_files

```
db_user: admin
db_password: admin@123
```

— —> now Run the playbook

ansible-playbook vault-playbook.yml --ask-vault-pass

Enter the vault password you used during encryption.

```
root@ip-172-31-1-201:~/playbooks# ansible-playbook vault-playbook.yml --ask-vault-pass
Vault password:

PLAY [Inject Ansible Vault variables] *****

TASK [Use vault variables] *****
ok: [localhost] => {
  "msg": [
    "DB User is admin",
    "DB Password is admin@123"
  ]
}

PLAY RECAP *****
localhost                : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

root@ip-172-31-1-201:~/playbooks#
```

i-0e47def07407ab494 (ansible-controller)

Ansible Vault – Task Explanation

- Created a vault file to store sensitive variables.
- Encrypted the vault file using `ansible-vault encrypt`.
- Injected vault variables into playbook using `vars_files`.
- Executed playbook by providing vault password securely.

Common Vault Commands

```
ansible-vault view vault.yml
ansible-vault edit vault.yml
ansible-vault decrypt vault.yml
ansible-vault rekey vault.yml
```

Conclusion:

Ansible Vault is used to encrypt sensitive variables and inject them securely into playbooks using `vars_files`.

